# ENGINEERING MECHANICS

# &

# SMART BUILDINGS

# CIE 1073

# LECTURE - 0

# COURSE CONTENT

1. Resultant of concurrent and non-concurrent coplanar forces
2. Equilibrium of concurrent and non-concurrent coplanar forces
3. Smart systems in building
4. BIM, AR and VR
5. Simple stresses and strains
6. Measurements using sensors
7. Case studies on IoT and cloud computing

# REFERENCE BOOKS


1. Engineering Mechanics - Meriam & Craige, John Wiley & Sons.
2. Engineering Mechanics - Irwing Shames, Prentice Hall of India.
3. Mechanics for Engineers - Beer and Johnston, McGraw Hills Edition
4. Strength of materials - Beer and Johnston
5. Strength of materials - F L Singer & Andrew Pytel
6. Strength of Materials - S S Bhavikatti
7. Smart building systems for Architects, owners and builders -James Sinopli Elsevier, Butterworth-Heinemann Publications, 2010
8. BIM Handbook: A Guide to Building Information Modeling for Owners, Managers, Designers, Engineers and Contractors- Eastman, C; Teicholz, P; Sacks, R; Liston, K, NewYork: Wiley. 2011
9. Introduction to Measurements & Instrumentation - A.K.Ghosh, IIIrd, PHI

# LECTURE - 1

- Concept of Rigid body
- Force and its characteristics
- Principle of transmissibility
- Classification of Force System
- Resultant of Coplanar concurrent forces
- Composition of forces
- Resolution of a force
- Rectangular Components of a force
- Sign convention

# INTRODUCTION TO RIGID BODY MECHANICS

## Rigid Body

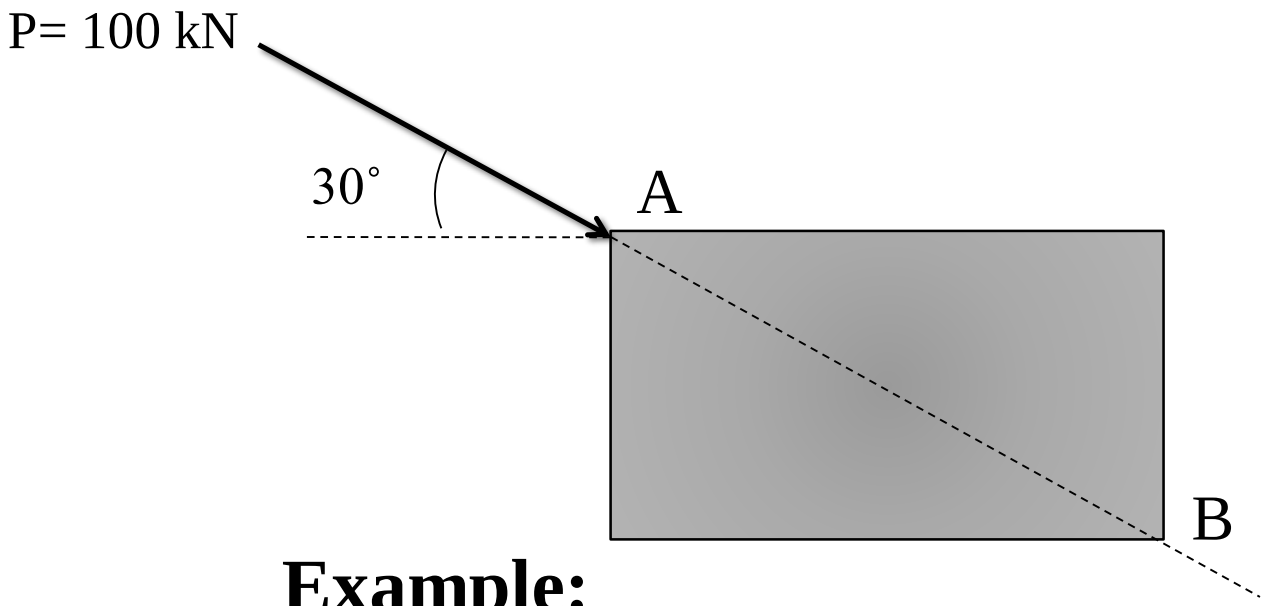
- A body is said to be rigid if it does not get deformed under the action of external load.
- It is defined as a definite amount of matter, the parts of which are fixed in position relative to one another.
- Actually solid bodies are never rigid; they deform under the action of applied forces. In those cases where this deformation is negligible compared to the size of the body, the body may be considered to be rigid.

# Force

- It is defined as the action of one body on another. (OR) It is that agent which causes or tends to cause changes or tends to change the state of rest or motion of a mass.

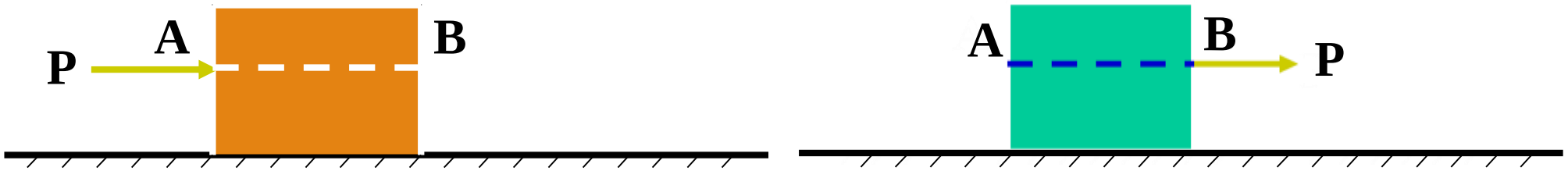
- Characteristics of a force:

  1. Magnitude
  2. Direction
  3. Point of application and
  4. Line of action




**Example:**

# Principle of Transmissibility

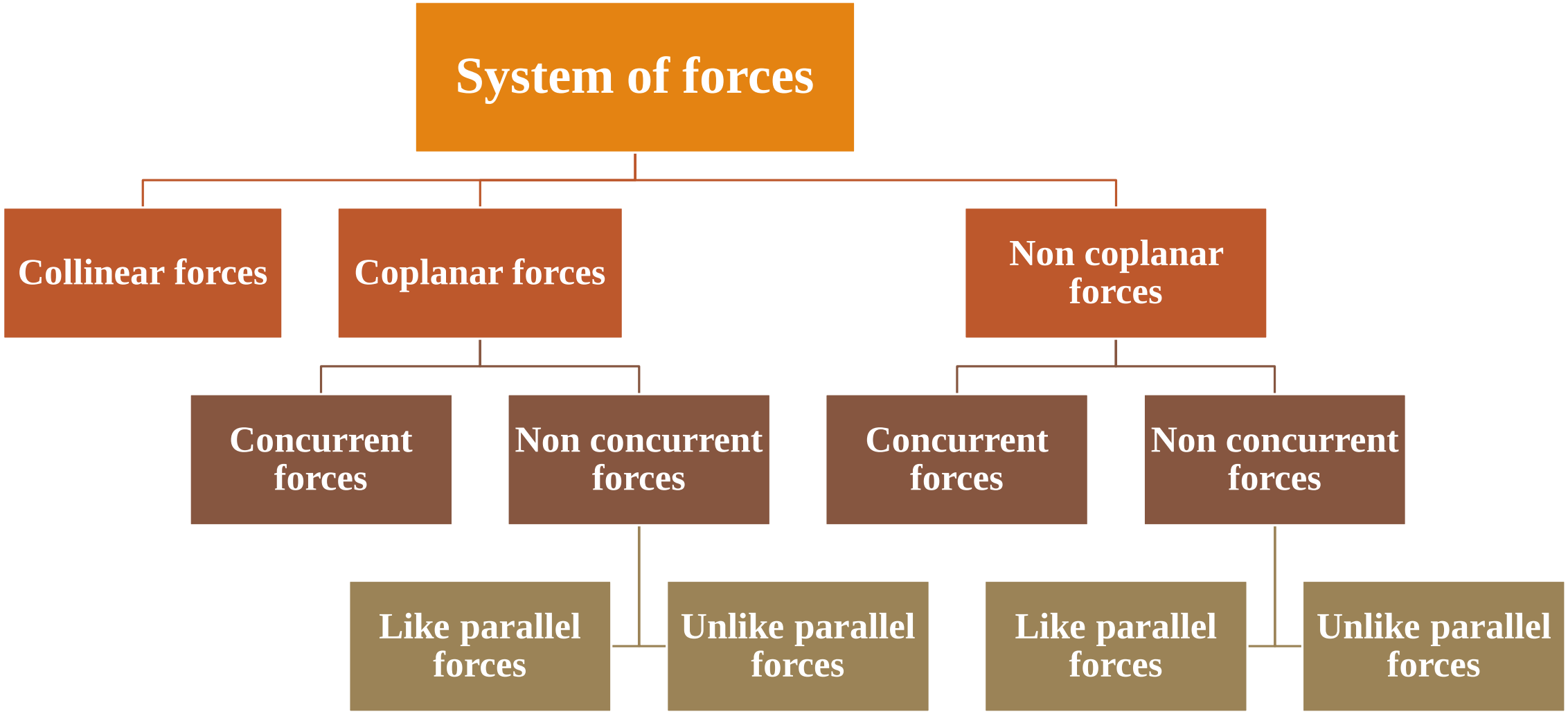



- The state of rest or motion of the block will be the same if a force of magnitude P is applied as a push at A or as a pull at B.

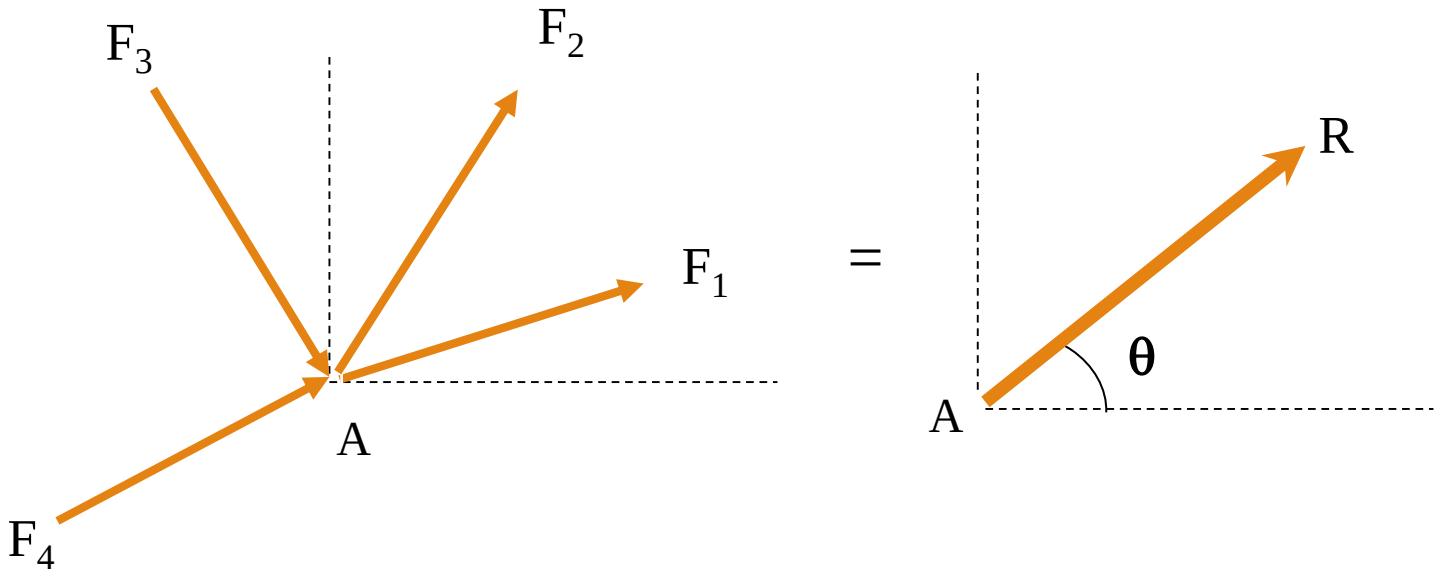
OR

- The state of rest or of uniform motion of a rigid body is unaltered if a force acting on the body is replaced by another force of the same magnitude and direction but acting anywhere on the body along the line of action of the replaced force.

# Classification of Force System



# RESULTANT OF COPLANAR CONCURRENT FORCE SYSTEM



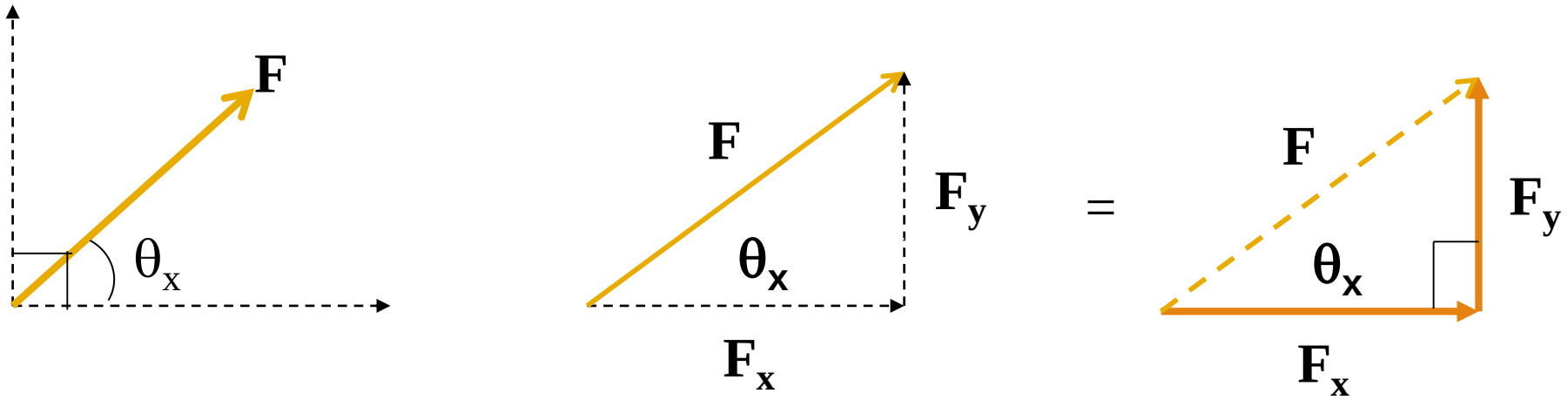

It is defined as that single force which can replace a set of forces, in a force system, and cause the same external effect.

- **Composition of forces:** It is the process of obtaining the resultant of a given force system.

- **Component of a force :** In simple terms, it is the effect of a force in a certain direction.

- **Resolution of a force:** The process of obtaining the components of a force.

# Rectangular Components of a force



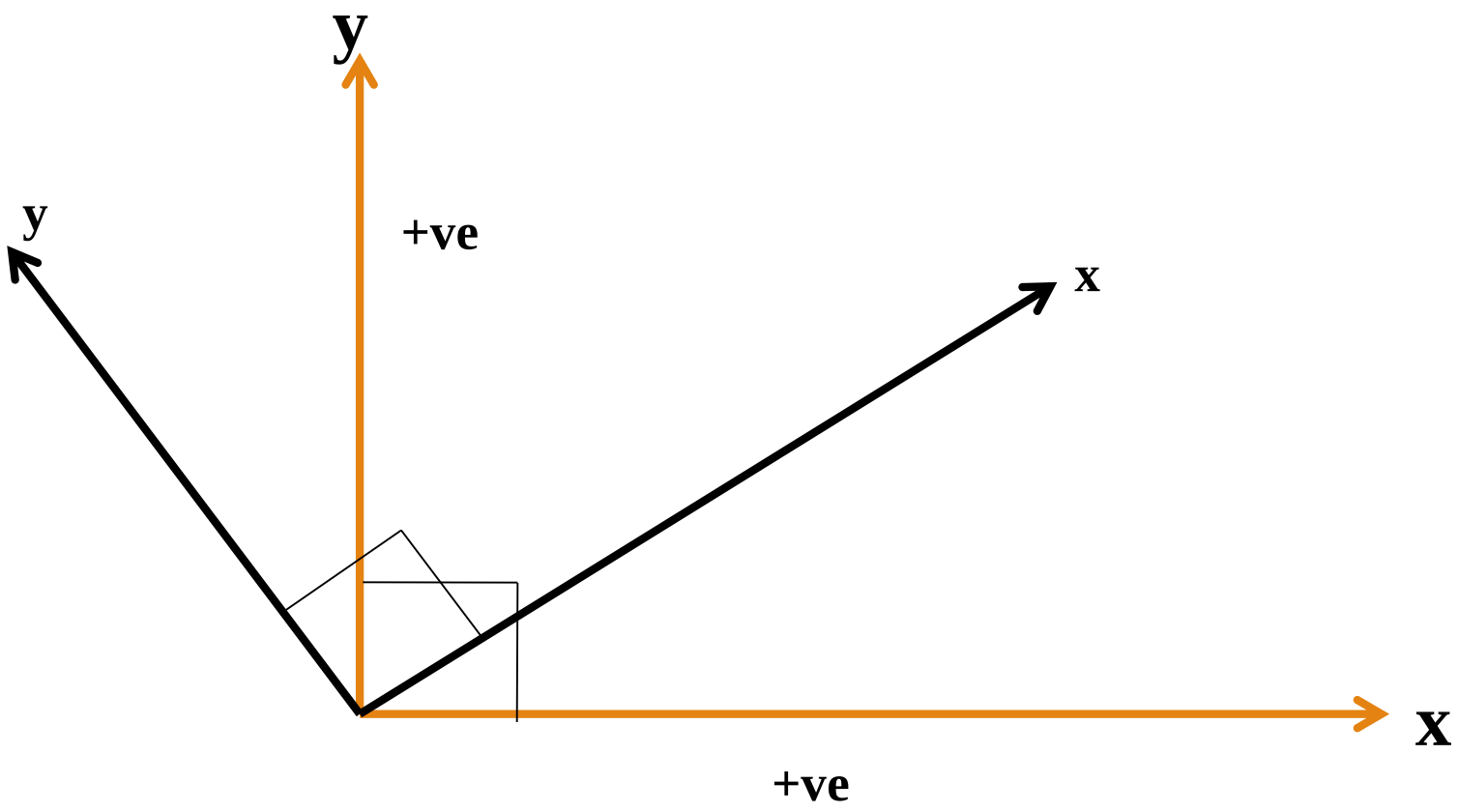

- The components that are mutually perpendicular are called '**Rectangular Components**'.

- Consider a force F making an angle $\theta_x$ with x-axis.

  Resolved part of the force F along x-axis is given by $F_x = F \cos \theta_x$

  Resolved part of the force F along y-axis is given by $F_y = F \sin \theta_x$

# Sign convention



# LECTURE - 2

- Resultant of Coplanar non-concurrent force systems
- Moment of a force
- Varignon’s theorem
- Couple
- Moment of a couple
- Resolution of a force into a force and couple
- Properties of a couple

# RESULTANT OF COPLANAR NON-CONCURRENT FORCE SYSTEM

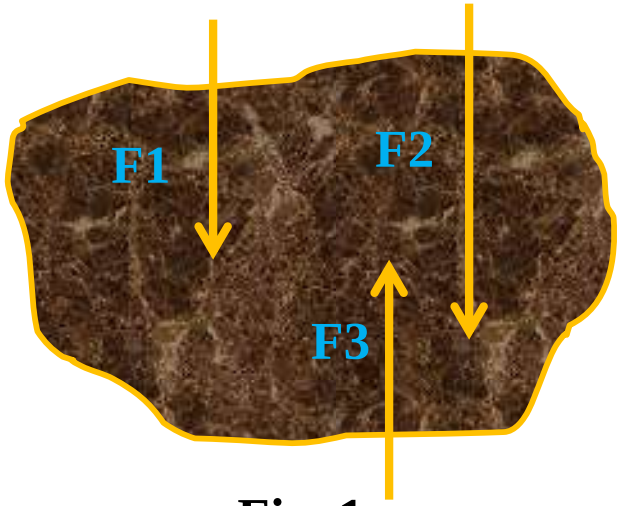



**Fig. 1**

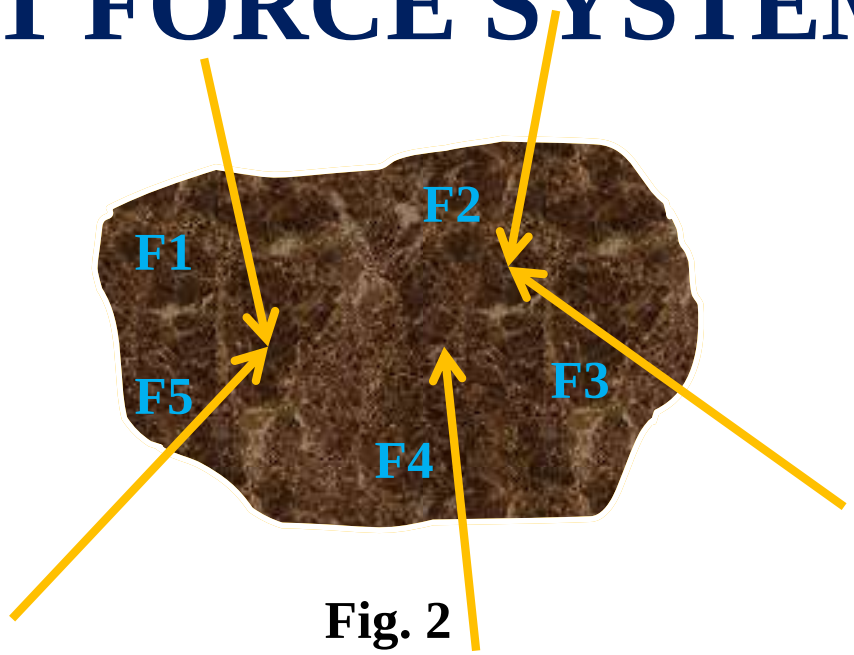
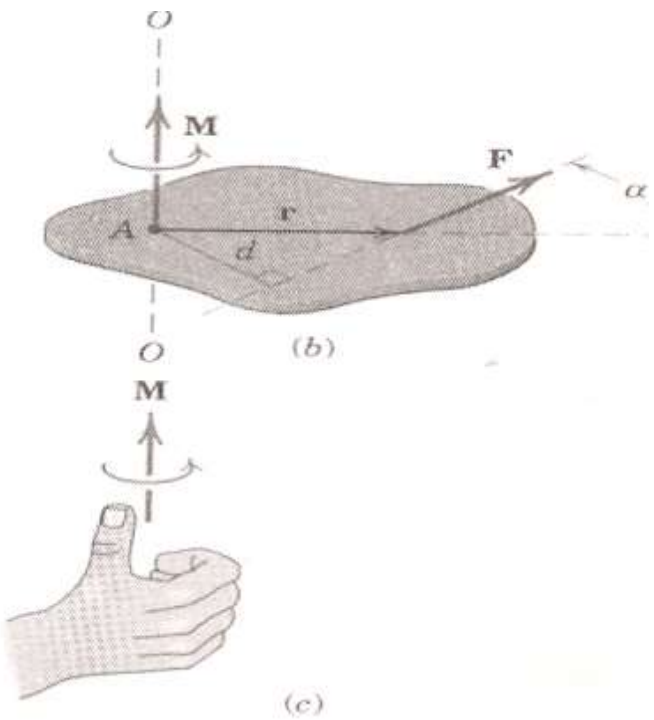



**Fig. 2**

- **Parallel Force System –** Lines of action of individual forces are parallel to each other.
- **Non-Parallel Force System –** Lines of action of the forces are not parallel to each other.

# Moment

- Moment of a force about an axis:
  - Moment Axis: the axis about which rotational tendency is determined.
  - Moment Center: This is the position of axis on co-planar system.
  - Moment Arm: Perpendicular distance from the line of action of the force to moment center.




- Magnitude of moment:

  $M_A = F \times d$

  = Rotation effect because of the force F, about the point A (about an axis *0-0*)

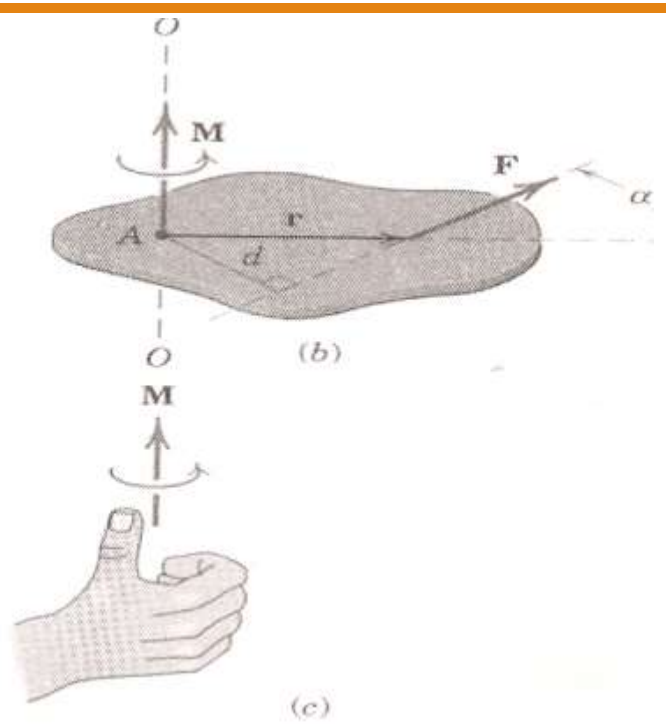
  Unit – kN-m, N-mm etc.

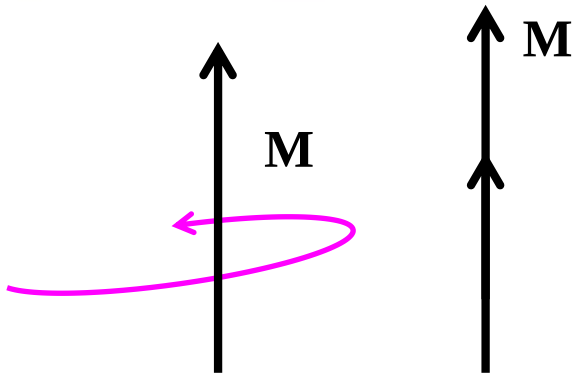
# Sense of a Moment

‘Right Hand Thumb’ rule**:**

‘If the fingers of the right hand are curled in the direction of rotational tendency of the body, the extended thumb represents the sense of moment vector’.

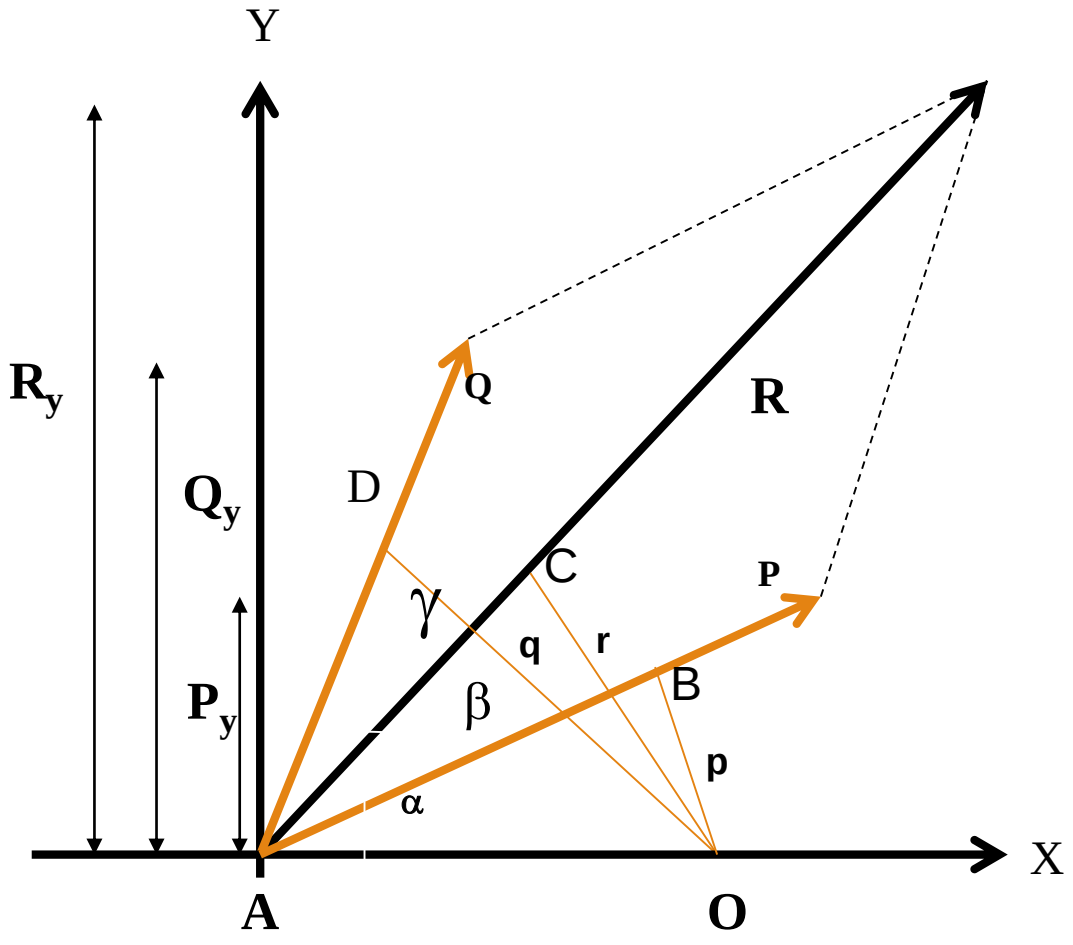
For the purpose of additions, the moment direction may be considered by using a suitable sign convention such as +ve for counterclockwise and –ve for clockwise rotations or vice-versa.



# Varignon's Theorem (Principle of moments)

Proof (by Scalar Formulation):



Let ‘R’ be the given force. ‘P’ & ‘Q’ are component forces of ‘R’.‘O’ is the moment center. p, r and q are moment arms from ‘O’ of P, R and Q respectively.$\alpha$, $\beta$ and $\gamma$ are the inclinations of ‘P’, ‘R’ and ‘Q’ respectively w.r.to X – axis.

We have,

$R_y = P_y + Q_y$

$R \sin\beta = P \sin\alpha + Q \sin\gamma$ ----(1)

From $\Delta$le AOB, $p/AO = \sin\alpha$

From $\Delta$le AOC, $r/AO = \sin\beta$

From $\Delta$le AOD, $q/AO = \sin\gamma$

From (1),

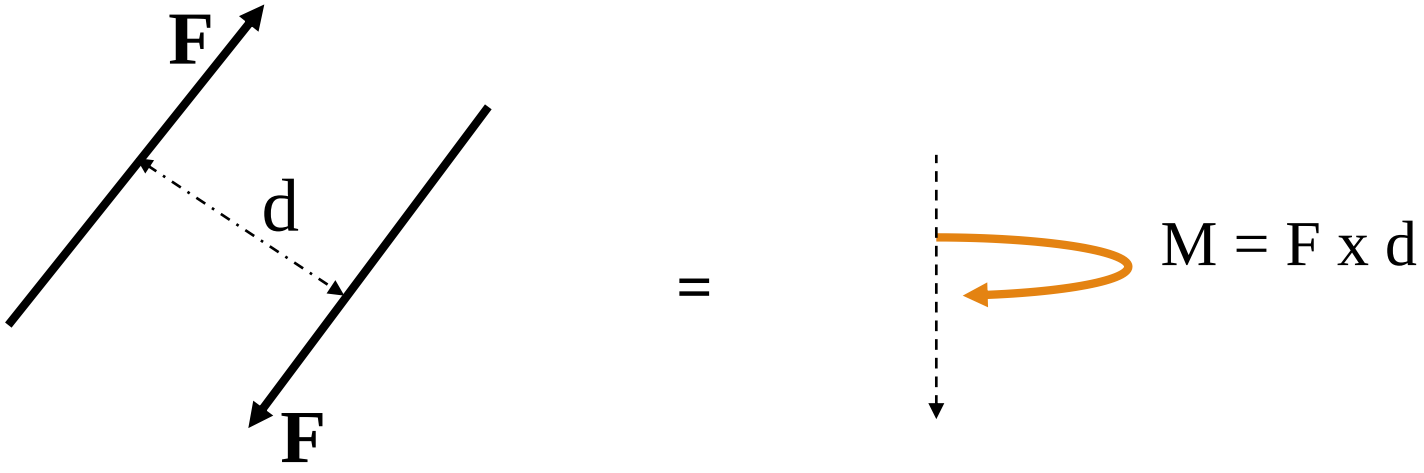
$$R \times (r/AO) = P \times (p/AO) + Q \times (q/AO)$$

i.e., $R \times r = P \times p + Q \times q$

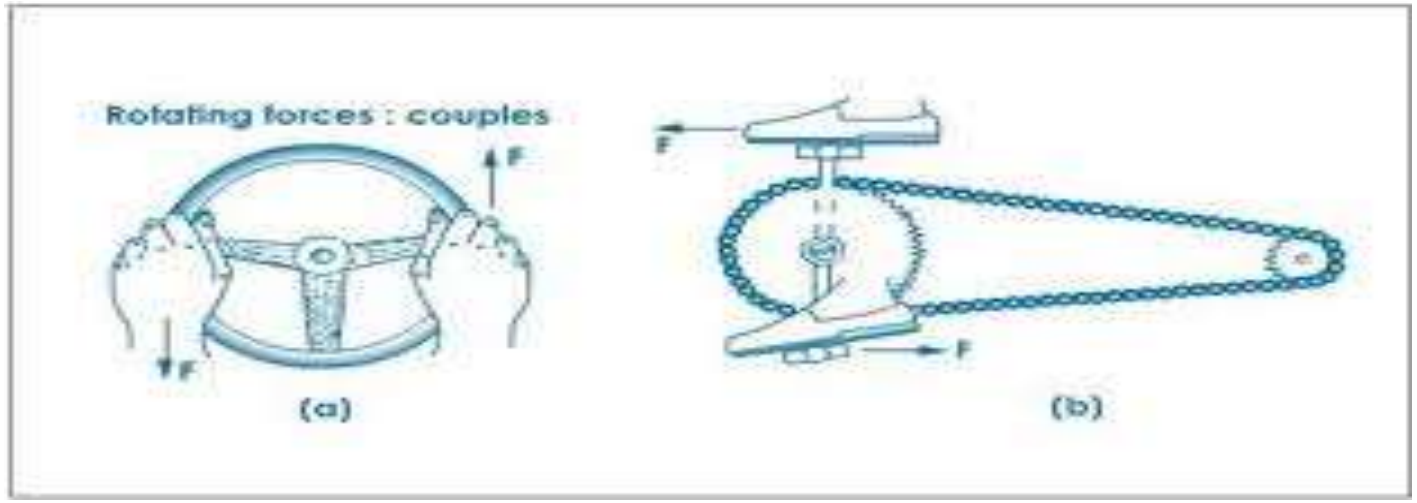
Moment of resultant R about O = algebraic sum of moments of component forces P & Q about same moment center ‘O’.

# Couple

Two parallel, non collinear (separated by certain distance) forces that are equal in magnitude and opposite in direction form ‘couple’.

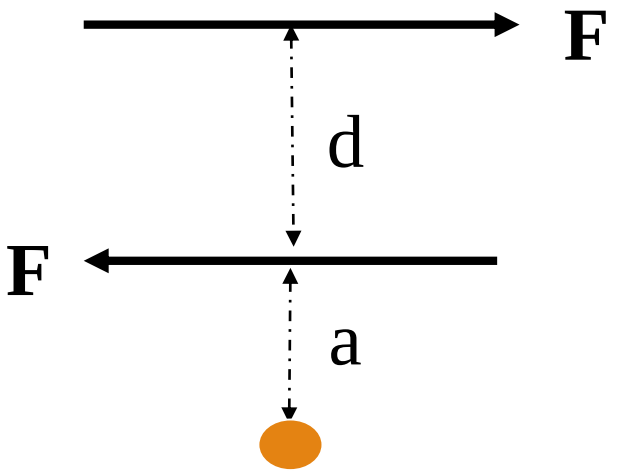



**<u>Couple does not produce any translation but produces only rotation.</u>**


Rotating forces : couples
F
F
(a)
F
F
(b)

A
-F
B
F
C
L/2
L/2

# Moment of a Couple




- Consider two equal and opposite forces separated by a distance ‘d’. Let ‘O’ be the moment center at a distance ‘a’ from one of the forces.
- The sum of moments of two forces about the point ‘O’ is,
  $\sum Mo = F \times (a + d) - F \times a = F \times d$

Thus, the moment of the couple about ‘O’ is independent of the location as it is independent of ‘a’.

# Properties of a Couple

- The algebraic sum of the components of the two forces is zero. i.e. the resultant force of a couple is zero
- The moment of a couple is constant for any point chosen in the plane of the couple
- A couple can be balanced by an equal and opposite couple in the same plane.
- Two or more couples can be reduced to a single couple of moment equal to the algebraic sum of the moments of the given couples
- The moment of a couple is independent of the choice of the axis of moments (moment centre). The moment of a couple is the same with respect to any axis perpendicular to the plane of the couple.

# Properties of a Couple cont…

- The effect of a couple is unchanged if the couple is rotated through any angle.
- Since the only effect of a couple is a moment and this moment is the same about any point, the effect of a couple is unchanged if :

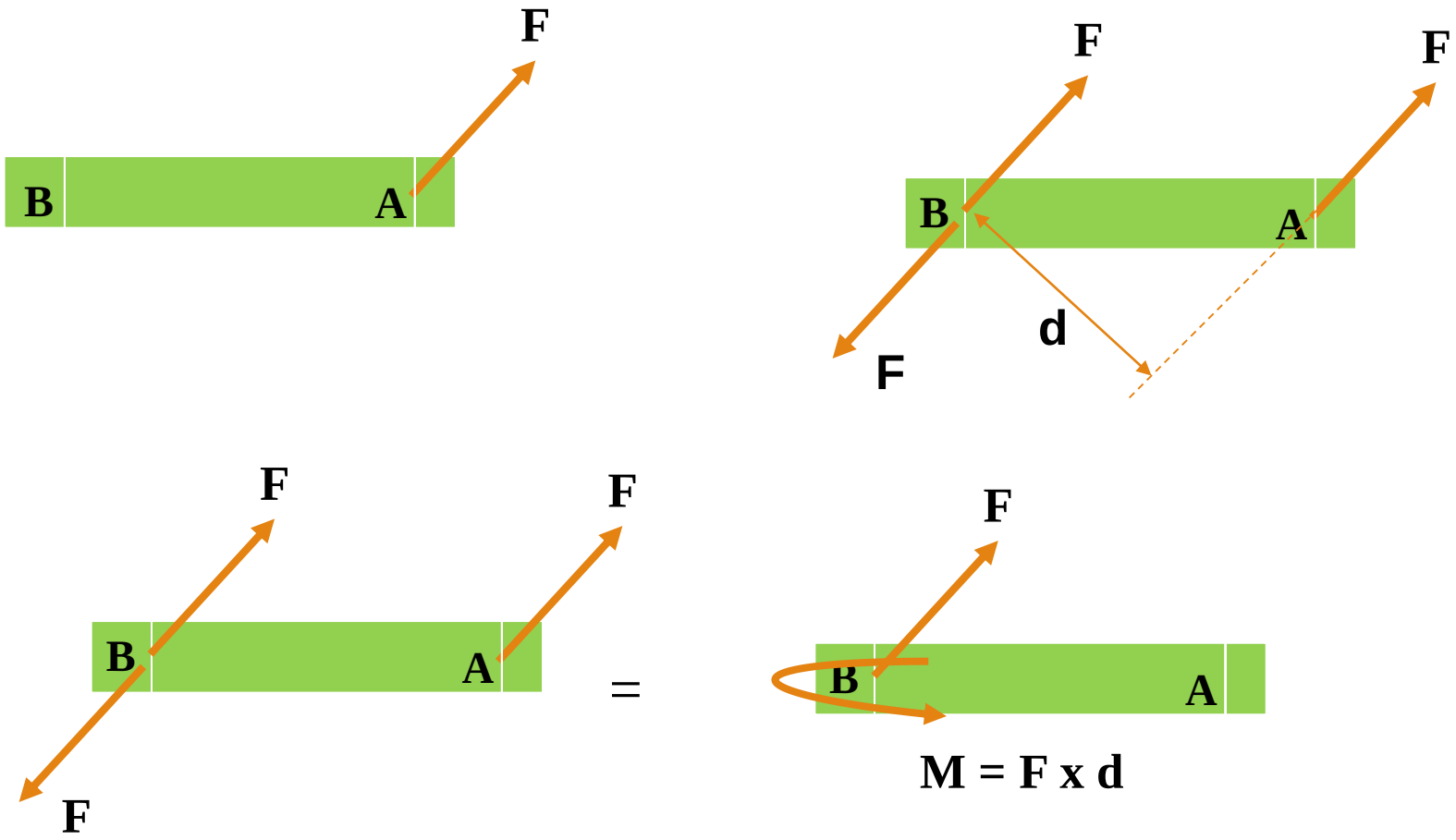
  The couple is rotated though any angle.

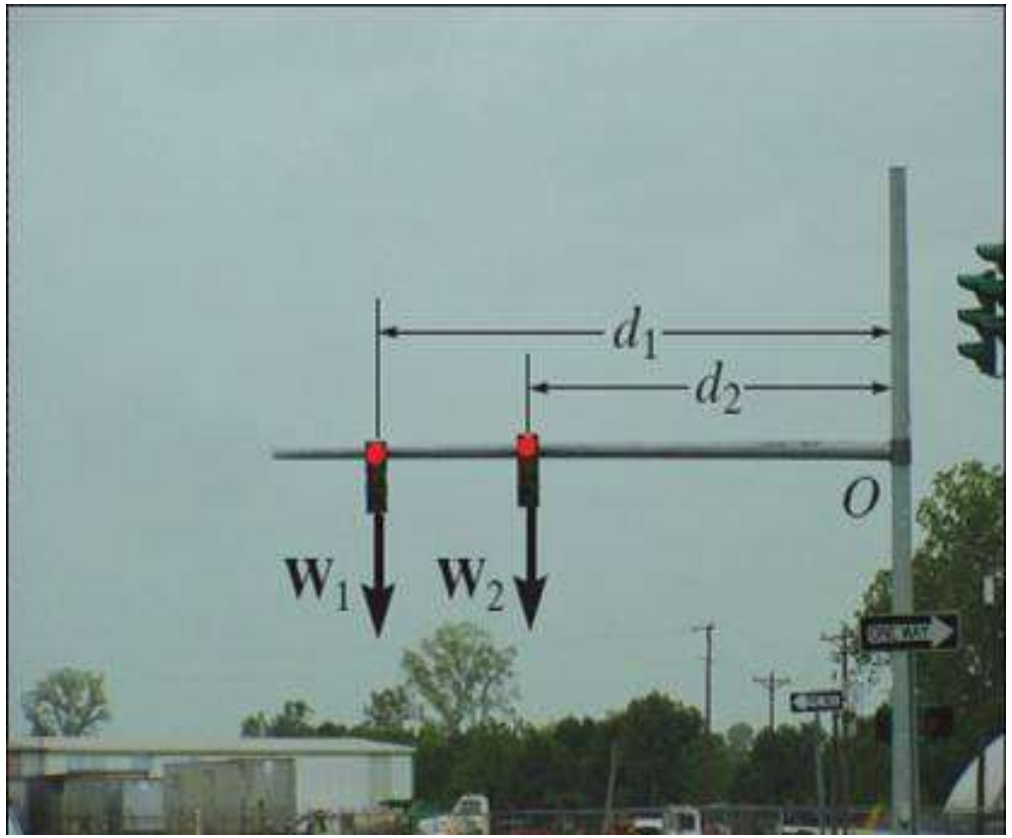
  The couple is shifted to any other position.

  The couple is replaced by another pair of forces whose rotational effect is the same

# Resolution of a force into a force-couple system

To replace the force F acting at the point A to the point B



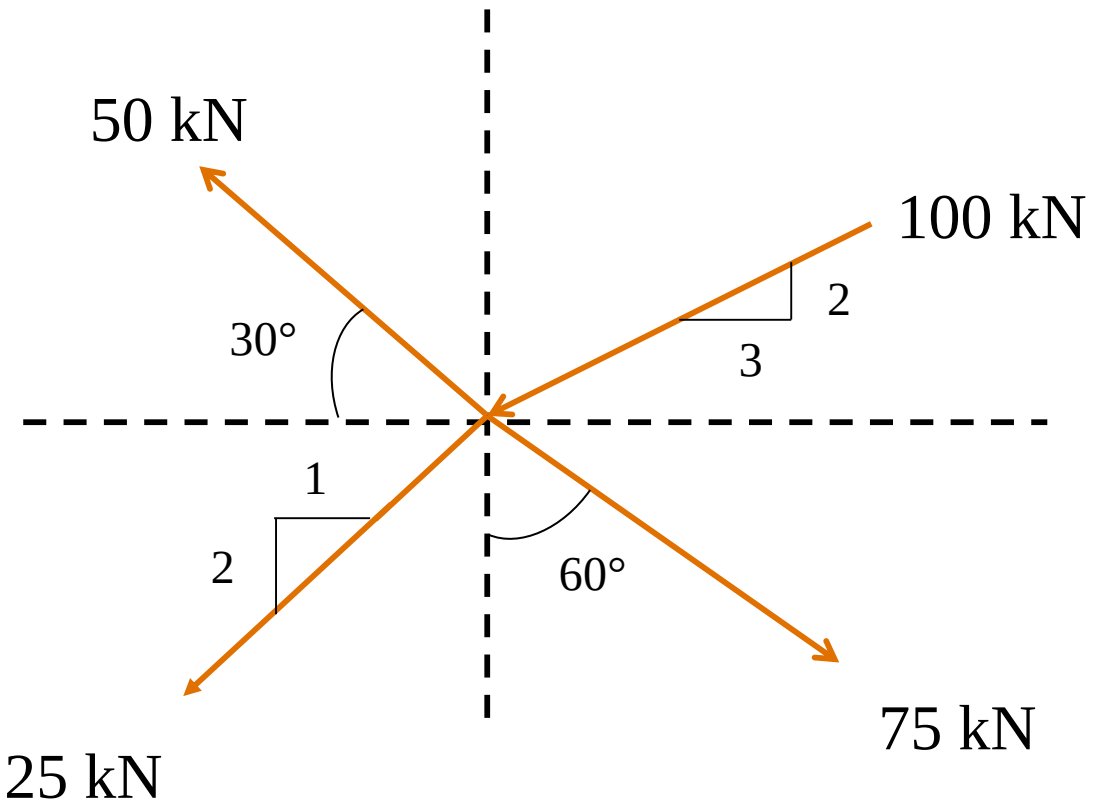

$d_1$
$d_2$
$O$
$\mathbf{W}_1$
$\mathbf{W}_2$

$(\mathbf{M}_R)_O$
$O$
$\mathbf{W}_R$

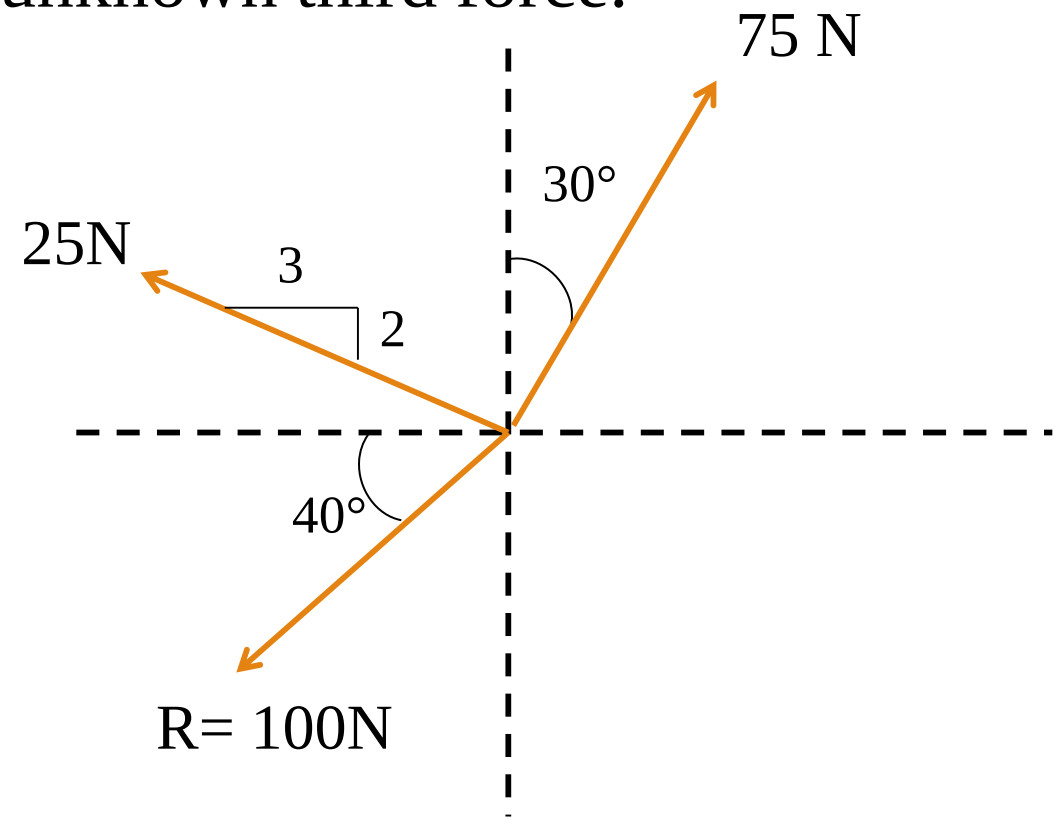
# LECTURE - 3

- Application Problems

1 . Obtain the resultant of the concurrent coplanar forces acting as shown in figure

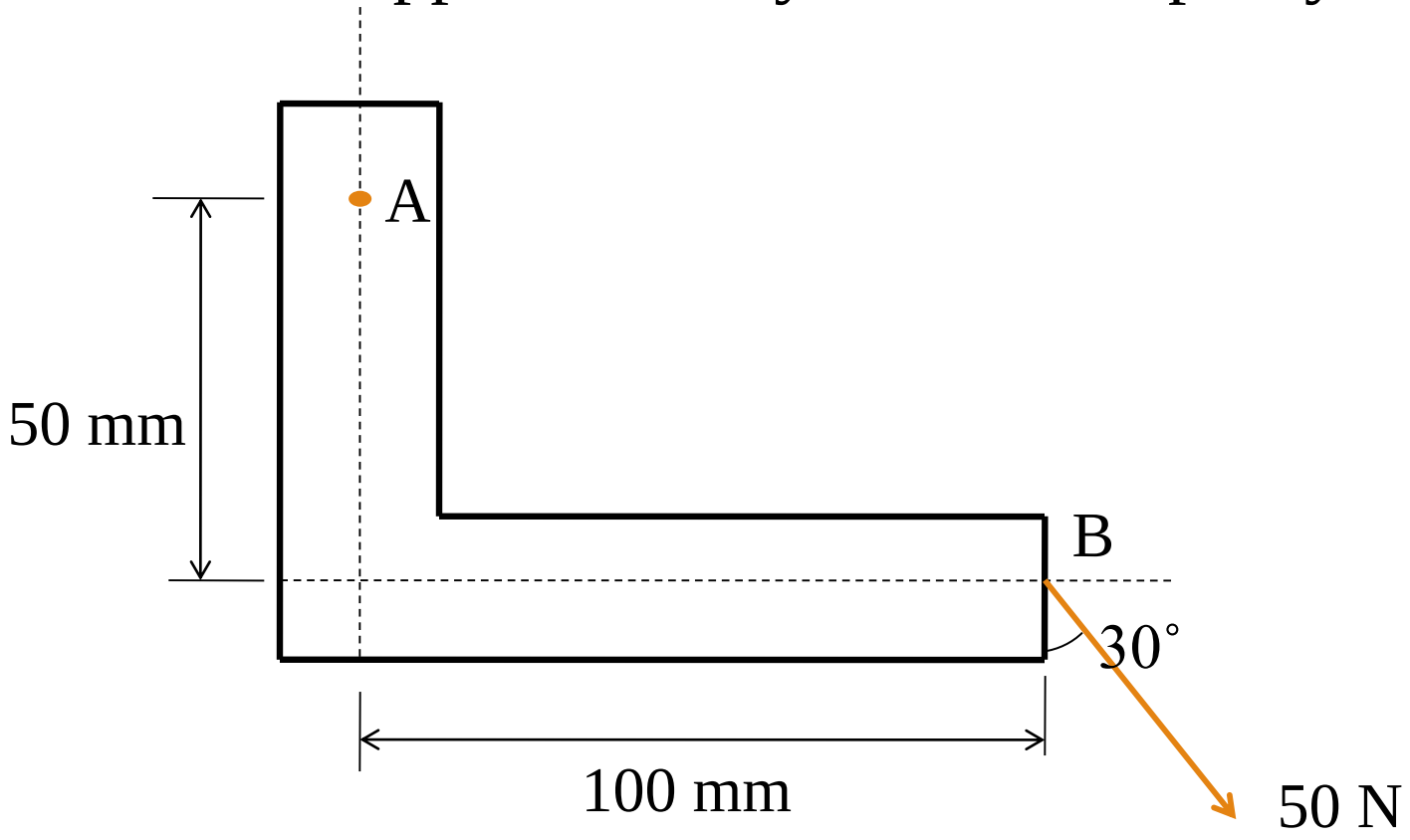


2. A system of concurrent coplanar forces has three forces of which only two forces are shown in the Fig. If the resultant is a force R = 100N acting as indicated, obtain the unknown third force.

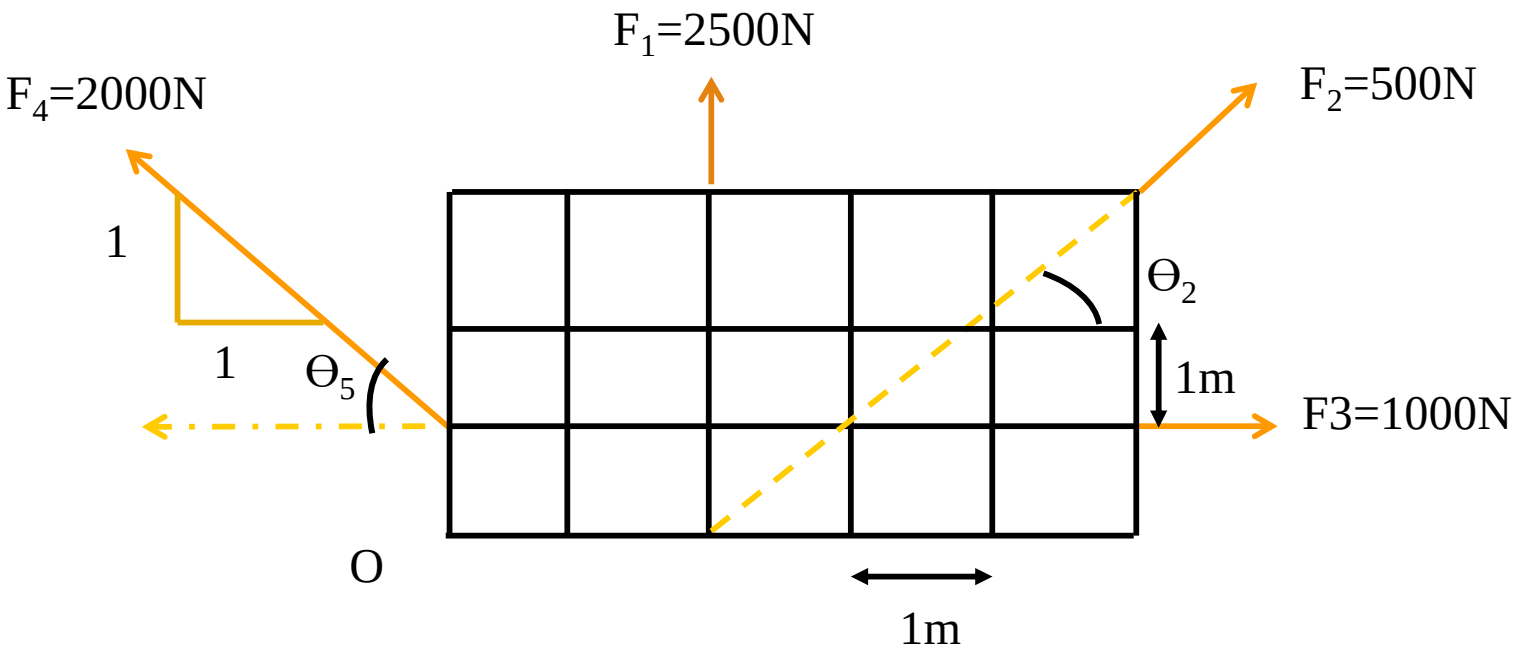


# LECTURE - 4

3. Replace the 50 N force applied at B by a force-couple system acting at ‘A’.

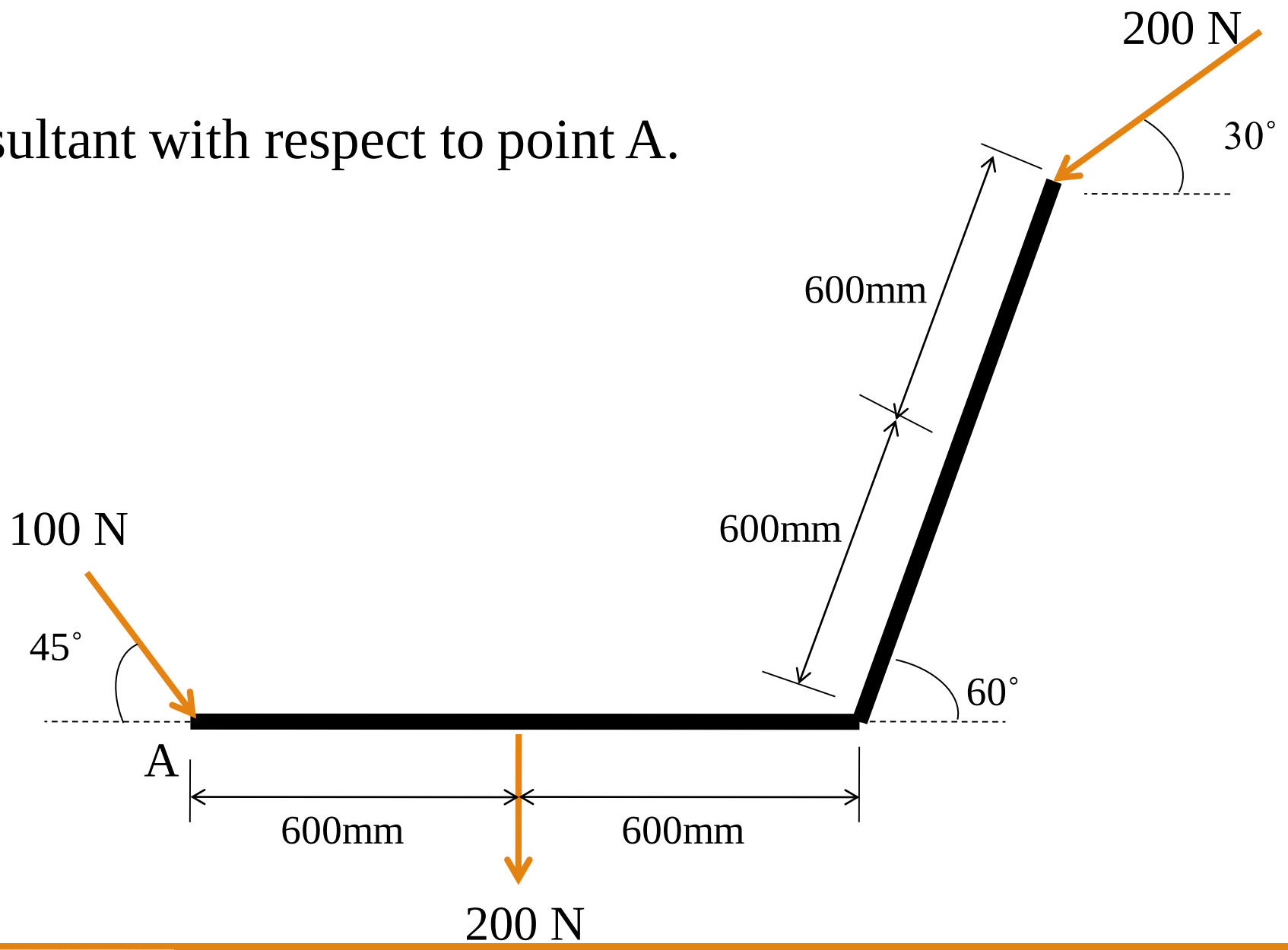


4. Find the resultant and its position w.r.to ‘O’ of the non-concurrent system of forces shown in the figure.

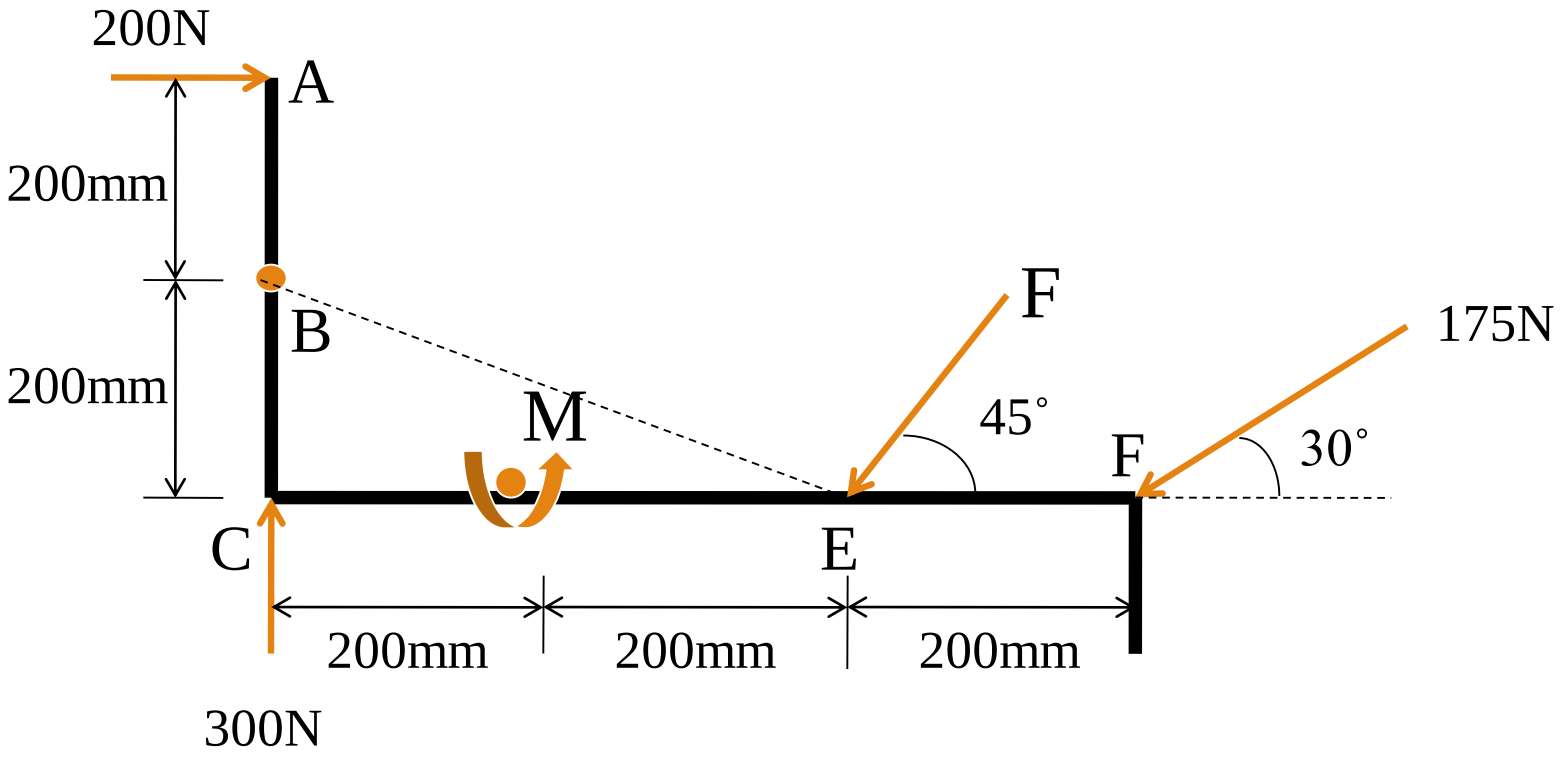


# TUTORIAL- 1

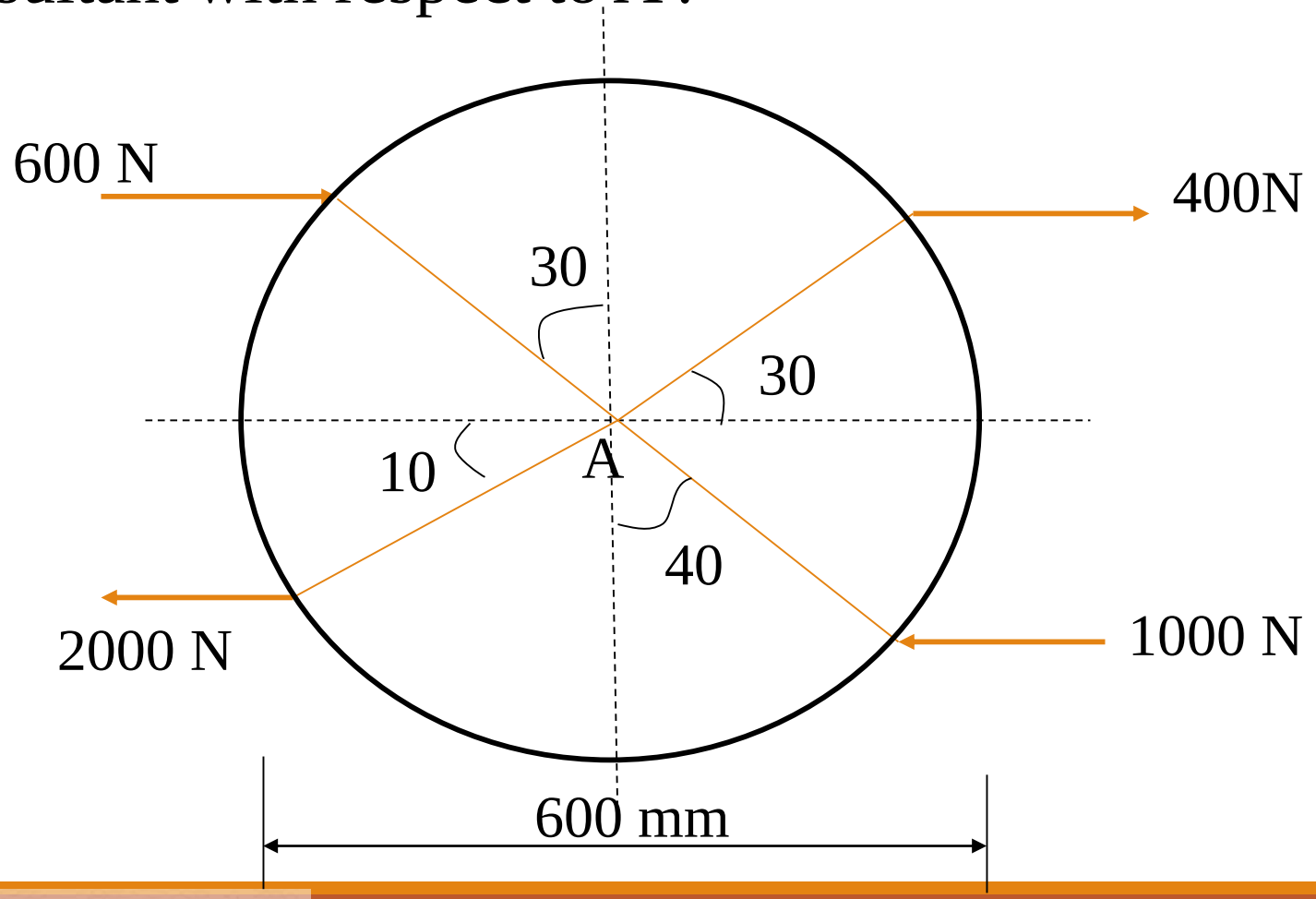
1. Locate the resultant with respect to point A.



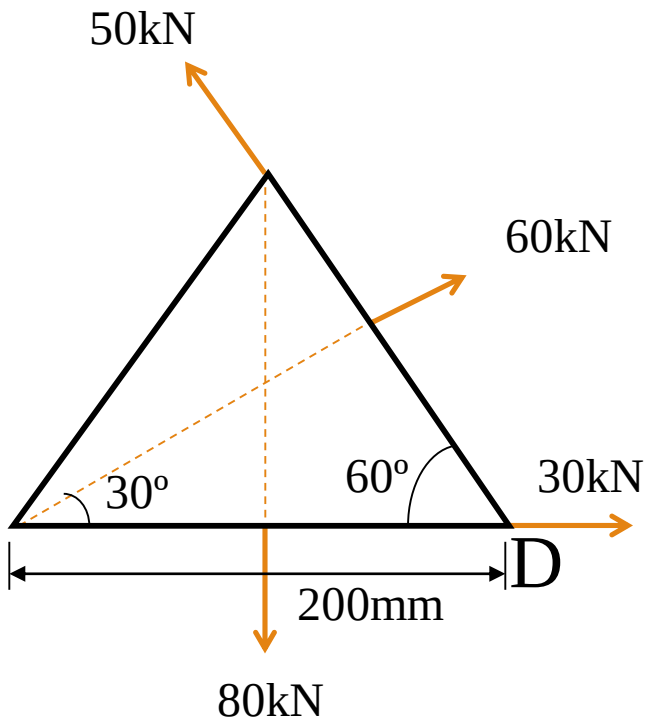
2. For the configuration shown in the figure, resultant of the force system acting on the frame passes through the point B and E as shown in the figure. Find the values of F and M.



3. Determine the resultant of the parallel coplanar force system shown in fig. Locate the resultant with respect to A .

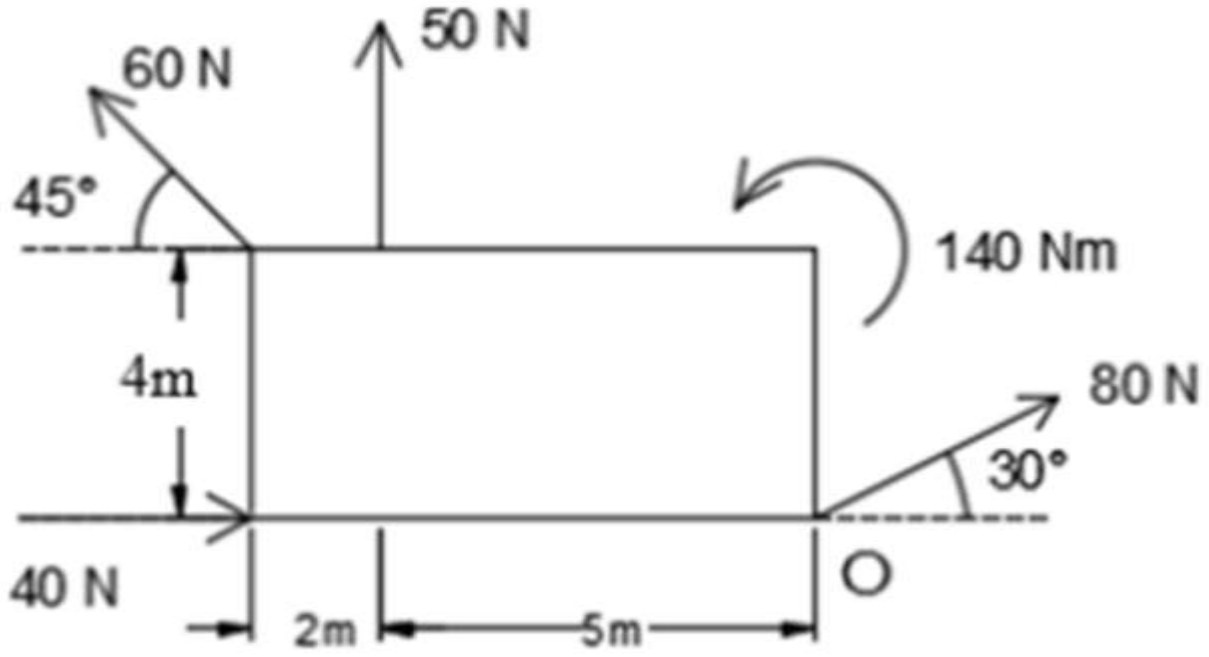


4. An equilateral triangle of sides 200mm is acted upon by 4 forces as shown in the figure. Determine magnitude and direction of the resultant and its position from point ‘D’.



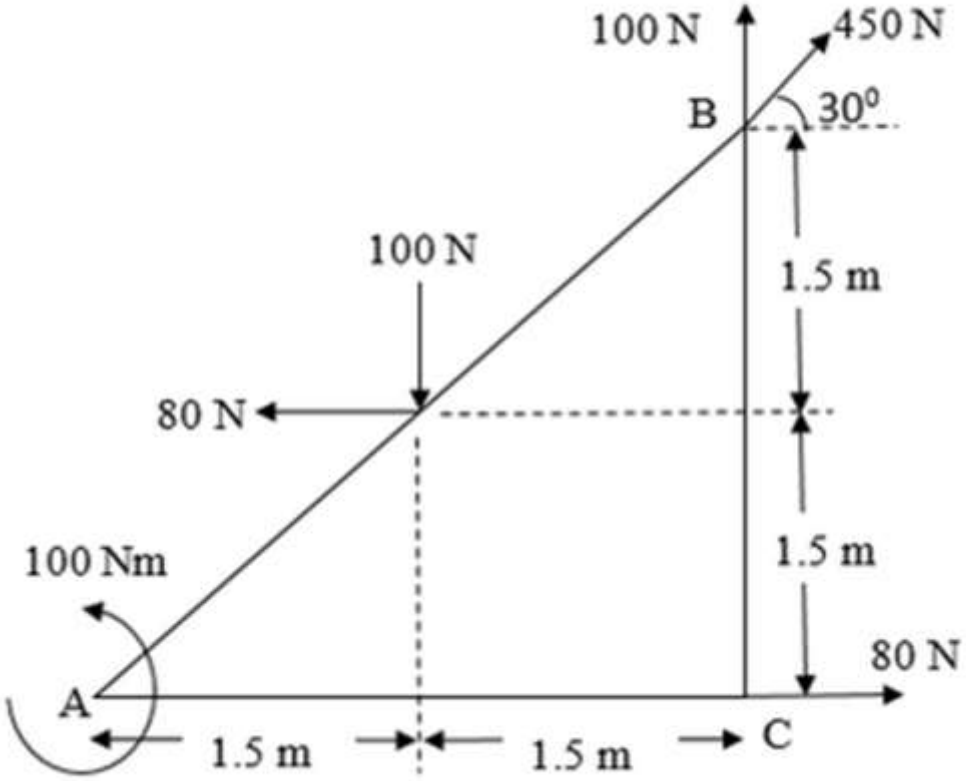

HOME

Find magnitude, direction and position of a resultant force for a system of forces shown in the figure with respect to 'O'.




R=148.33 N
d= 3.48 m

Locate the resultant of a force system shown in the figure with respect to C.




R=449.99 N
d= 1.77 m